Empiric anti-anaerobic antibiotics are associated with adverse clinical outcomes in emergency department patients

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To the Editor:

The protective effects of commensal, anaerobic,gut microbiota have been established in various acute diseases [1–3]. This raises the question whether antibiotics depleting these anaerobic bacteria come with unexpected side-effects. We, therefore, read with great interest the article by Chanderraj *et al.* [4], which identified associations between treatment with anti-anaerobic antibiotics and adverse clinical outcomes in 3032 mechanically ventilated patients. By characterising gut microbiota in 116 patients and through animal experiments, the authors found evidence of causal effects of anti-anaerobic antibiotics on infectious and non-infectious outcomes. These findings demonstrate the harmful effects of anaerobic microbiota depletion in intensive care unit (ICU) patients. Yet, antibiotic treatment policies significantly differ between institutes and countries, and the protective effect of gut microbiota might extend beyond ICU patients. We thus wondered if anti-anaerobic antibiotics were also associated with adverse clinical outcomes in a different country and a broader population that includes all patients admitted through the emergency department (ED). Here, we aimed to validate and expand on the findings by Chanderraj *et al.* [4] in electronic health record data of ED patients admitted to Amsterdam University Medical Center (UMC) in the Netherlands.

We intended to mimic the approach taken by Chanderraj *et al.* [4], with minor alterations to increase relevance for our study population. Ventilator-associated pneumonia is a less relevant outcome in ED patients. We, therefore, focused on mortality and nosocomial infection-free survival (defined as no positive cultures obtained at least 48 h after admission). Like Chanderraj *et al.* [4], multivariable Cox proportional hazard models were built with anti-anaerobic antibiotics, hospital admission unit, age, sex and Charlson Comorbidity Index. We included the modified-early warning score (MEWS) [5, 6] rather than APACHE IV to quantify disease severity, which is more suitable for an ED population. Missing MEWSs were handled using mean-imputation together with an indicator variable (indicating per patient whether MEWS was measured), as described [7]. The Amsterdam UMC local medical ethics review committee waived the review of this retrospective cohort study (IRB number: IRB00002991; case: 19.499), as the medical research involving Human Subjects Act did not apply. De-identified data extracts were used for this study, adhering to the local privacy officer's protocol. Therefore, no informed consent needed to be obtained for the use of the data.

We identified 15 908 adult patients visiting our ED between 2016 and 2020 who were admitted to the hospital and treated with systemic antibiotics. Of 15 908 patients, 80.2% (n=12 755) received "anti-anaerobic antibiotics" (as defined by Chanderraj *et al.* [4]) in the first 24 h following ED arrival, which is higher than the 64.1% in the original study. In our cohort, patients who received anti-anaerobic antibiotics were older (mean 60.5±18.1 *versus* 56.4±19.1 years; p<0.0001), equally often male (57.5% *versus* 58.1%; p=0.58), and had a higher MEWS (1.11 *versus* 0.68; p<0.0001) and Charlson Comorbidity Index (2.35 *versus* 1.98; p<0.0001) than patients who did not receive anti-anaerobic antibiotics in the first 24 h. According to the manuscript's classification [4], ceftriaxone (33.5%; 7716 of 23 016 unique prescriptions) and amoxicillin with clavulanic acid (15.3%; 3521 of 23 016) were the most frequently administered anti-anaerobic antibiotics. Gentamicin (9.9%; 2289 of 23 016) was the most administered antibiotic without anti-anaerobic activity.

In our cohort, exposure to anti-anaerobic antibiotics was associated with a higher risk of mortality at 30 days: 1263 deaths among 12 755 anti-anaerobic treated patients *versus* 206 deaths among 3153 patients not treated with anti-anaerobic antibiotics (hazard ratio (HR) 1.55, 95% CI 1.34–1.80; p<0.0001) (figure 1a).

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In a retrospective cohort of 15 908 emergency department patients, empiric use of anti-anaerobic antibiotics was associated with decreased overall and infection-free survival in patients with suspected infection in the emergency department https://bit.ly/3TRbE0F

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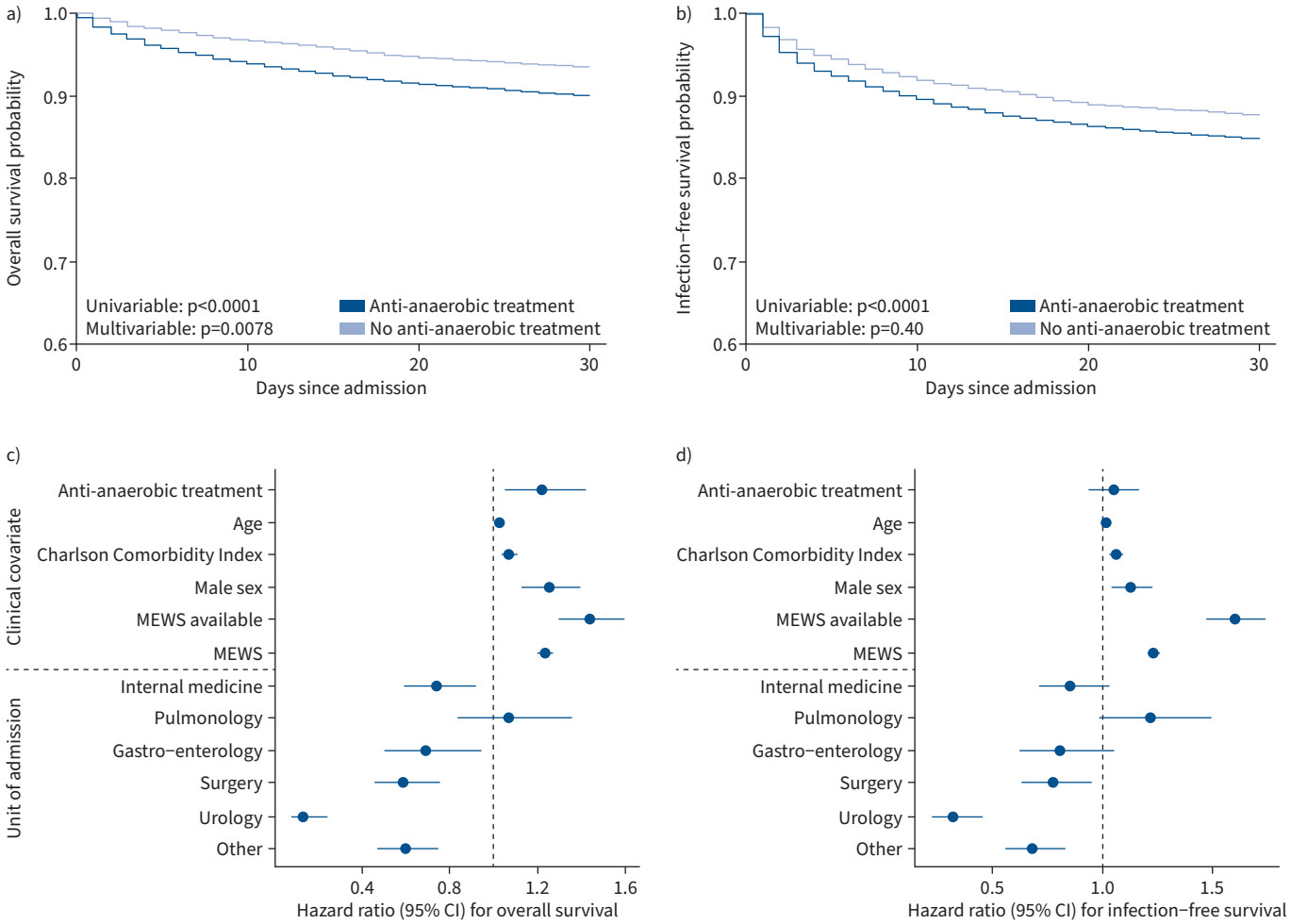


FIGURE 1 Anti-anaerobic antibiotics are associated with decreased survival and infection-free survival. In a cohort of 15 908 adults patients admitted to the hospital through the emergency department and treated with systemic antibiotics, the use of anti-anaerobic antibiotics on the first day was associated with decreased a) overall survival at 30 days (p<0.0001) and b) infection-free survival at 30 days in univariable analyses (p<0.0001). In a multivariable Cox regression model including anti-anaerobic antibiotics, hospital admission unit (neurology served as reference), age, sex, Charlson Comorbidity Index and disease severity (quantified by modified-early warning score (MEWS), and a categorical variable indicating per patient whether MEWS was measured to address data missing not at random), anti-anaerobic treatment was independently associated with an increased hazard of c) mortality at 30 days (p=0.0078). d) However, anti-anaerobic antibiotics were not independently associated with infection-free survival (p=0.40).

Similar to Chanderraj *et al.* [4], we corrected for demographics, MEWS, Charlson Comorbidity Index and admission unit in a multivariable model, and found that anti-anaerobic treatment was independently associated with an increased hazard of mortality (HR 1.22, 95% CI 1.05–1.42; p=0.0078) (figure 1b). Although anti-anaerobic antibiotics were associated with reduced infection-free survival in univariable analysis (HR 1.27, 95% CI 1.13–1.41; p<0.0001) (figure 1c), this was not observed after multivariable adjustment (HR 1.05, 95% CI 0.94–1.17; p=0.40) (figure 1d). No interactions between treatment effects and disease severity or department were observed.

In our setting, not all antibiotics classified as "anti-anaerobic" by Chanderraj *et al.* [4] are necessarily prescribed for anaerobic coverage [8]. We thus performed sensitivity analyses in which only piperacillin–tazobactam, meropenem, metronidazole, clindamycin, and amoxicillin with clavulanic acid were categorised as anti-anaerobic antibiotics. Exposure to these more strictly defined anti-anaerobic antibiotics (n=5653 patients) was associated with decreased overall (HR 1.16, 95% CI 1.03–1.29; p=0.010) and infection-free survival (HR 1.14, 95% CI 1.04–1.24; p=0.0039) in multivariable analyses. We thus concluded that anti-anaerobic antibiotic treatment was associated with adverse outcomes, not only in ICU patients, but also in our broader population, and when using a stricter definition of anti-anaerobic antibiotics.

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Similar to those of Chanderraj *et al.* [4], these observational study findings are vulnerable to bias by indication and unmeasured confounding. Despite the well-established protective effects of anaerobic microbiota in animal experiments [1–4, 9], human interventional studies are warranted.

In this retrospective cohort of ED patients, we found associations between anti-anaerobic treatment and adverse outcomes, as also described by Chanderraj *et al.* [4], regardless of differences in study populations, antibiotic exposure and outcome rates. We validated and extended the observations in mechanically ventilated patients to a broader population. Taken together, these findings suggest that we should not only reconsider the widespread use of anti-anaerobic antibiotics in the ICU, but in all hospitalised patients.

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Conflict of interest: The authors have nothing to disclose.

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